

Final Project Report

Statistical Analysis and Predictive Modeling of the Ames Housing
Dataset

Stephen Codoluto

Purdue University
MSDS GRAD50500

February 2026

1. Project Introduction

This report provides an analysis of the Ames Housing Dataset¹. This dataset was originally compiled by Dean De Cock from the Ames City Assessor's Office and published in the Journal of Statistics Education in 2011 (De Cock, 2011). It contains 82 variables that include building types, lot information, quality ratings, square footage, room counts, garage capacity, basement features, and sale information. A detailed overview of the dataset is provided in **Table 1**.

The primary objective of this project is to apply hypothesis testing and statistical analysis techniques to better understand the factors that drive home sale prices. This objective was framed using three research questions. Each question was evaluated using statistical methods to test hypotheses about correlations between home features and price, neighborhood effects, and median sale price predictions using home features.

The statistical methods utilized in this project include multiple linear regression with F-tests and individual predictor t-tests to determine coefficient impact, an ANOVA (**AN**alysis **Of** **VA**riance) F-test for a two-model comparison, logistic regression for binary classification, and a binomial Z-test to evaluate classification accuracy against random chance. A 95% confidence interval is also calculated for classification accuracy in the third research question.

The dataset csv file was obtained from <https://www.openintro.org/data/index.php?data=ames> and is attached as a supporting file in the Appendix.

Table 1: Dataset Overview

Meta-data	Description
File Format	Comma-Separated Values
Data Type	Tabular / Structured, 1 observation per row
Observations	400 random houses selected of 2,929 total available
Measurement Formats	Categorical and Numeric
Variables	82 columns (23 nominal, 23 ordinal, 14 discrete, 20 continuous, 2 identifiers)
Granularity	Individual residential property (one row per house sale)
Temporal Details	2006 – 2010 (sales dates)
Geographic Details	Ames, Iowa, USA - subdivided into neighborhoods

1. De Cock, Dean. "Ames, Iowa: Alternative to the Boston housing data as an end of semester regression project." Journal of Statistics Education 19.3 (2011).

2. Research Questions and Hypotheses

Research Question 1

What physical attributes of a home most strongly predict its sale price?

This question explores which measurable features have the strongest relationship with sale prices. The response variable is sales price, and the dependent/explanatory variables are the measurable features including living area, overall quality, basement size, garage capacity, and year built/remodeled.

H_0 : All regression coefficients equal 0.

H_1 : At least one regression coefficient is not equal to 0.

Statistical Methods: Multiple linear regression with F-test for overall model significance and individual t-tests for each coefficient. Significance level $\alpha = 0.05$.

Research Question 2

Does neighborhood significantly affect home sale prices with house size, quality, and other impactful variables controlled?

This question tests whether location/neighborhood significantly impacts sales prices beyond what physical features can explain. Two models are compared. The first model looks at physical features only. The second model adds neighborhood and compares the difference from the first model.

H_0 : Neighborhood has no effect on price after controlling for physical attributes.

H_1 : Neighborhood has a significant effect on price after controlling for physical attributes.

Statistical Methods: ANOVA F-test comparing linear regression models. Significance level $\alpha = 0.05$.

Research Question 3

Can it be predicted whether a house will sell above the median price based on its features?

This question addresses a practical scenario for a buyer who wants to know if a home is predicted to sell below or above median price for the market given a home's physical features. The response variable is binary (sale price above/below median). The explanatory variables are the top six most strongly correlated physical features.

H_0 : Model classification accuracy = 0.5.

H_1 : Model accuracy > 0.5 - better than random chance.

Statistical Methods: Logistic regression model to predict above/below median price, a binomial Z-test for classification accuracy, and 95% confidence interval for model accuracy. Significance level $\alpha = 0.05$.

3. Analysis

Exploratory Data Analysis

The dataset was first explored to understand the data cleaning and preparation required prior to being used in the analysis. A random sample of 400 homes was selected from the original dataset of 2929 houses to keep the dataset and associated computing manageable.

After looking at the shape and general structure of the dataset, each column was analyzed to count the number of null values shown in **Table 2** for the features with the highest number of null values.

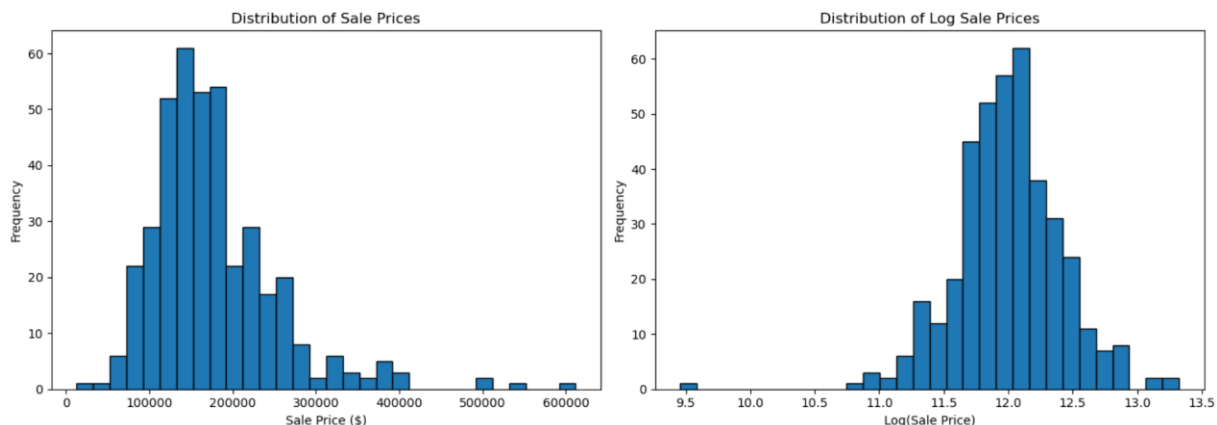
Table 2: Missing Values in Ames Dataset

Feature	# Nulls
Pool.QC	400
Alley	380
Misc.Feature	374
Fence	313
Mas.Vnr.Type	248
Fireplace.Qu	198
Lot.Frontage	71
Garage.Qual	24
Garage.Type	24
Garage.Finish	24

Several columns had a high proportion of missing data. Variables with highest missing rates like Pool.QC, Alley, and Misc.Feature represent amenities that do not exist on most properties. These types of variables were removed since it does not make sense to impute a feature that does not exist. Any columns with > 60% data missing were removed. The Order and PID columns were not needed for analysis and removed. Categorical columns that had several missing values were filled with 'None'. Numeric columns with missing data were filled with 0. Lastly, Lot.Frontage missing values were imputed with the median of the lot frontage. The resulting clean dataset had 400 houses, 75 variables plus an index, and zero null values.

Figure 1 (Left) displays a histogram of sale prices in the Ames dataset and shows a price distribution that is right skewed with a calculated skewness of 1.70. Most homes are in the \$100k - \$200k range with a smaller fraction of homes stretching the right side in the \$200k - \$400k range and an even smaller percentage beyond \$400k. These higher end homes are inflating the upper end of the distribution. **Figure 1 (Right)** shows the natural log transformation of the data which results in a closer to normal distribution. The median sale price is \$160k with a mean price of \$177k and a standard deviation of \$75k. The higher mean relative to the median confirms the right skewness.

Figure 1: Distribution of Sale Prices (Left) and Log Sale Prices (Right)



This exploratory analysis and the visualizations helped determine how to approach the research questions and use of regression models further in the analysis. It also validated the approach in research question three to use a median approach versus mean.

Research Question 1

What physical attributes of a home most strongly predict its sale price?

Correlation Analysis

Correlation coefficients were calculated between all numeric variables and sale price. **Table 3** and **Figure 2** display a correlation heatmap of the top ten features. The far-right column and bottom row show the ranking displayed in **Table 3**, with Overall.Qual showing the highest correlation against price. Lighter shade cells indicate strong positive correlations, while darker red cells show weak to no relationships. This color-coding was helpful to quickly identify patterns between features and price.

The heatmap also provides insight into inter-column patterns. An example of this is the relationship between area and TotRms.AbvGrd which are correlated at 0.80. This makes sense since it should

be expected that there are typically more rooms with larger homes. Another example is between Garage.Cars and Garage.Area which are correlated at 0.89, demonstrating they are measures of similar features. The strong correlation with garage size could be a result of higher demand for garage space in colder climates. X1st.Flr.SF and Total.Bsmt.SF also show a strong correlation of 0.82, meaning first-floor footprint and basement area increase/decrease together as one would expect with homes that have basements. Year.Built and Year.Remod.Add are moderately correlated (0.59), likely because homes that have never been remodeled retain their original build year as the remodel year in the dataset.

As a result of the insights from the correlations between Garage.Cars - Garage.Area, X1st.Flr.SF - Total.Bsmt.SF, and area - TotRms.AbvGrd, Garage.Area, X1st.Flr.SF, and TotRms.AbvGrd were excluded from the regression models to reduce redundancy. Full.Bath was also excluded from the model due to the low correlation factor. The correlation heatmap was a useful visual in feature selection and helped to ensure the regression model captured distinct features.

Figure 2: Correlation Heatmap of Top 10 Sale Price Predictors

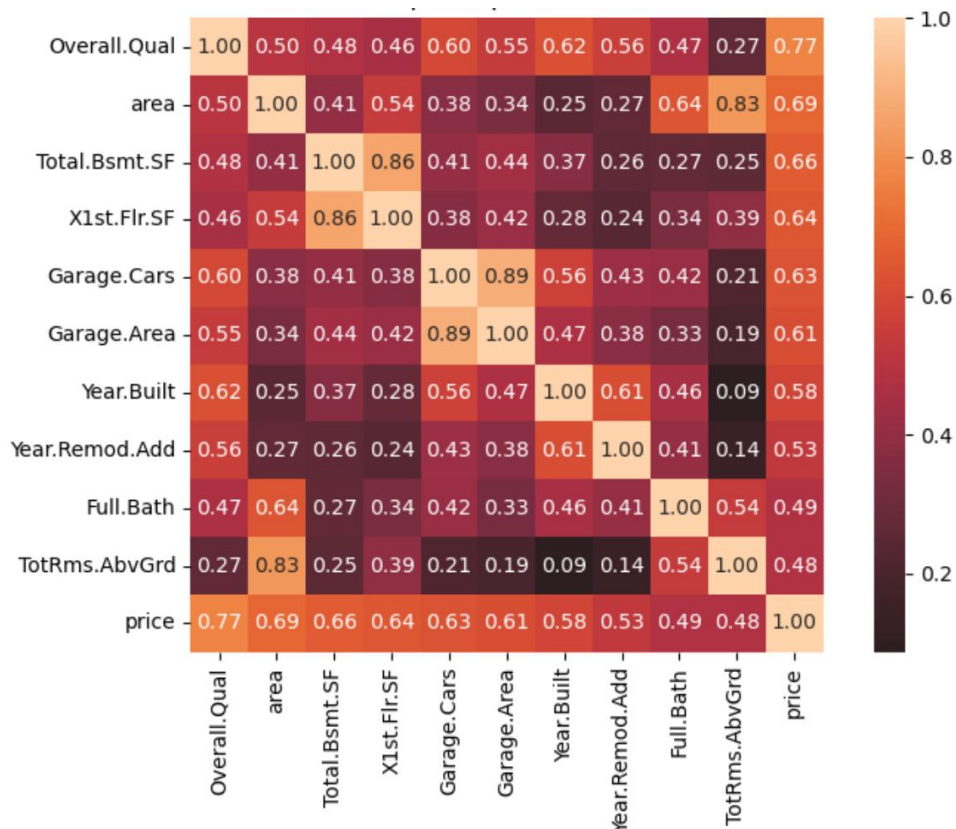


Table 3: Top 10 Correlated Physical Attributes w/ Sale Price

Physical Attributes	Correlation
Overall.Qual (Overall Quality)	0.77
Area (Total Living Area)	0.69
Total.Bsmt.SF (Basement Area)	0.66
X1st.Flr.SF (First Floor Area)	0.64*
Garage.Cars (Garage Capacity)	0.63
Garage.Area (Garage Area)	0.61*
Year.Built (Year Built)	0.58
Year.Remod.Add (Year Remodeled)	0.53
Full.Bath (# of Full Bathrooms)	0.49*
TotRms.AbvGrd (Total Rooms)	0.48*

*Variables were measures of similar home features or had a low correlation factor.

These variables were not included in the subsequent analyses.

Multiple Linear Regression

The statistical method used for this question was a multiple linear regression with an F-test for overall model significance and individual t-tests for each coefficient. Predictors included overall quality, living area (sq ft), basement size (sq ft), garage capacity, year built, and year remodeled. The six physical features explain 82% of the variance in sale prices and all six of these predictors contribute to the model. **Table 4** shows the individual feature coefficients and statistics while **Table 5** summarizes the model results.

Table 4: Multiple Linear Regression Coefficients

Feature	Coefficient	Std. Error	t-stat	p-value
Overall.Qual	1.52×10^4	1932.31	7.84	<0.0001
area	54.65	4.22	12.94	<0.0001
Total.Bsmt.SF	50.91	4.71	10.82	<0.0001
Garage.Cars	1.47×10^4	3089.70	4.76	<0.0001
Year.Built	223.37	84.57	2.64	0.009
Year.Remod.Add	401.07	108.47	3.70	<0.001

Table 5: Multiple Linear Regression Model Summary

Statistic	Value
R^2	0.82 (82% of variance explained)
F-statistic	290 (significant)
p-value	< 0.0001

Assumptions Met and Verified

The assumptions of the regression model were verified through several figures outlined below:

- **Linearity (Figure 3):** Verified using scatter plots showing linear relationships between the physical feature predictors and sale price.
- **Normality (Figure 4):** Met and verified using Q-Q plot of residuals. The residuals were normally distributed.
- **Constant Variance (Figure 5):** Plotted residuals vs. y to show minimal fanning or funneling.
- **Mean of 0 (Figure 5):** Confirmed residuals were not uniformly above 0 or uniformly below 0 for any subsection.
- **Independence (Figure 5):** Plot of residuals vs. y did not indicate any clear patterns. Each home sale was verified as a unique home sale with no repeated or duplicate rows.
- **Outliers:** Although there were three or four outliers, the relative percentage of outliers did not significantly impact the results. This was verified by removing the outliers and rerunning the models to get similar results.

Figure 3: Scatter Plots of Physical Features vs. Sale Price

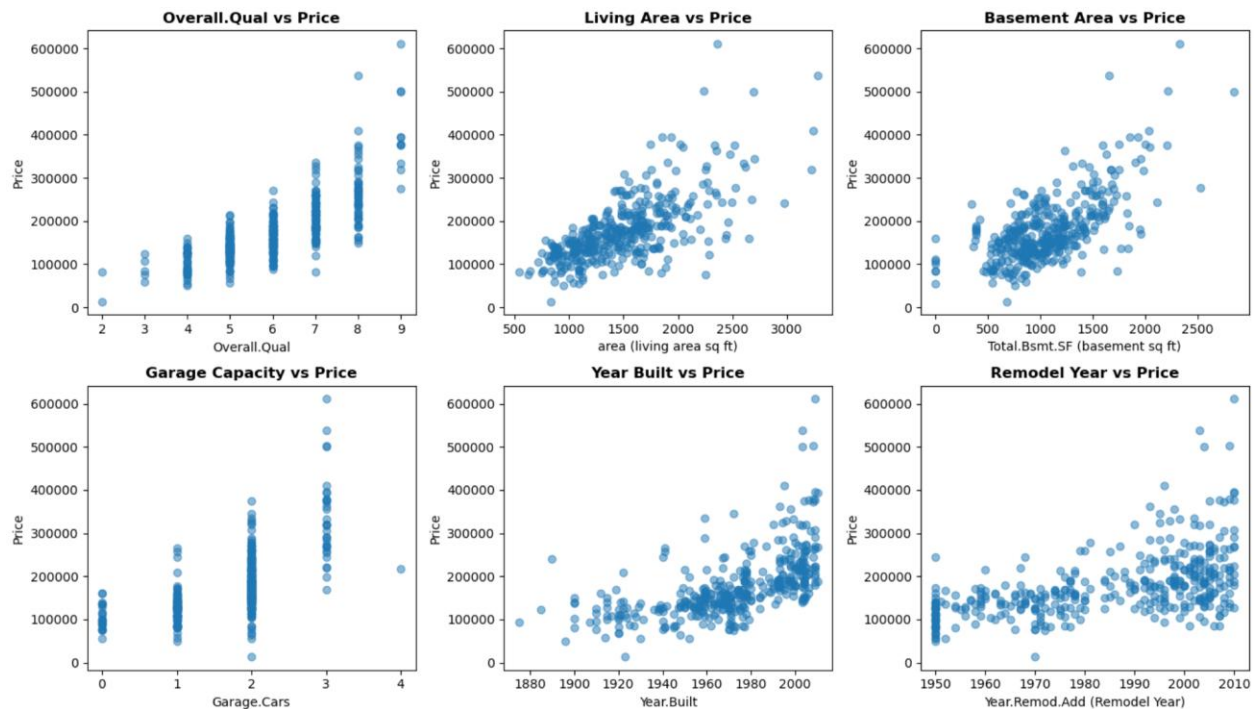


Figure 4: (Left) Q-Q Plot of Residuals and (Right) Residuals Distribution

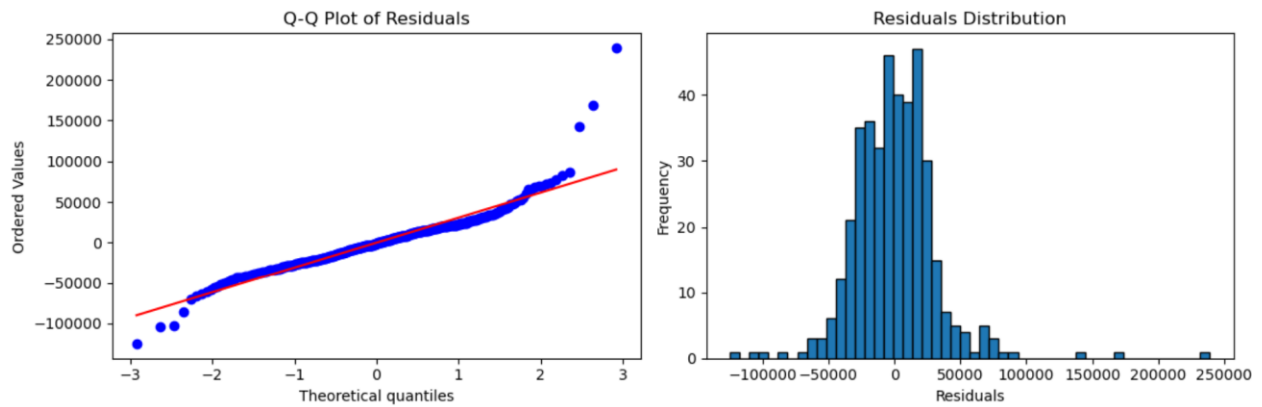
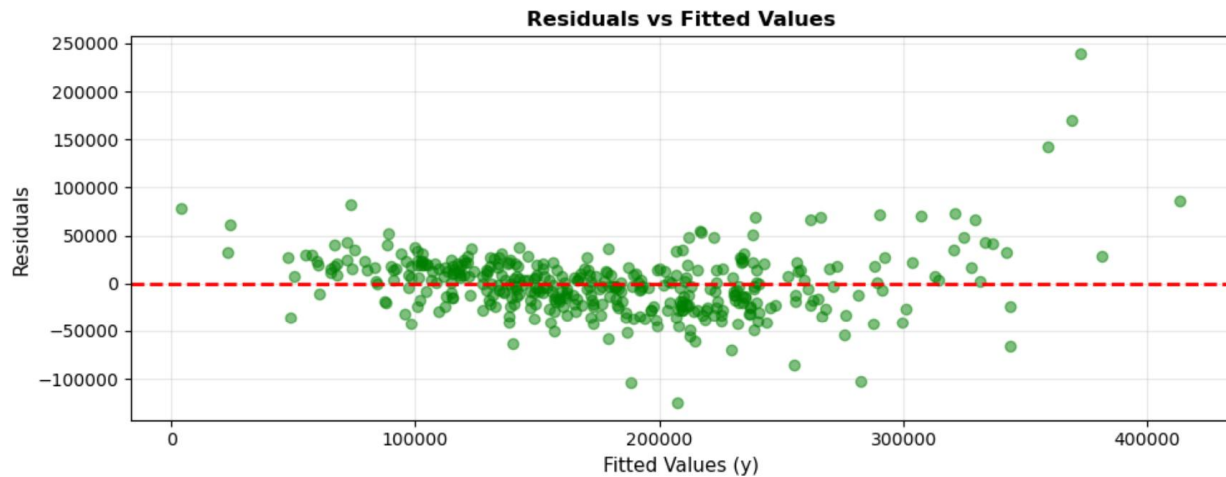


Figure 5: Constant Variance and Mean of 0: Residuals vs. Fitted



Based on the model results, at least one physical attribute predicts sale price, so we reject H_0 . There is sufficient statistical evidence that physical features significantly predict home sale prices. The F-test confirms the model is significant with $F = 290$ and $p < 0.0001$. The model explains 82% of the sale price variance.

Research Question 2

Does neighborhood significantly affect home sale prices with house size, quality, and other impactful variables controlled?

This question tests how impactful neighborhood location is after controlling for key physical features of a home. The statistical test used to analyze this relationship was an ANOVA F-test for a two-model comparison. Model 1 uses price against the top six correlated physical features and Model 2 uses price against the top six correlated physical features plus neighborhood names mapped to dummy variables for neighborhoods that had at least 5 houses in the dataset.

Table 6: ANOVA Model Comparison

Model	R ²	F Statistic	P-Value
Model 1 (Physical Features Only)	0.816		
Model 2 (Physical Features + Neighborhood)	0.841		
Delta R ²	0.025	3.015	0.000029

Based on the results outlined in **Table 6**, Neighborhood significantly affects sale price beyond physical attributes as calculated by the F-test showing a p-value of $0.000029 < 0.05$. Although statistically significant when using the F-test, the overall neighborhood effect relative to the top correlated physical features is small. Physical attributes remain the primary driver of price, explaining 82% of the variance.

Figure 6 shows the R² comparison between the models on the left. The middle and right sections of this figure show the top five neighborhoods with highest and lowest premiums, respectively. Neighborhood effects range from roughly -\$25k to +\$30k after controlling for physical features.

Figure 6: (Left) R² Comparison, (Middle) Top 5 Neighborhoods w/Highest Premium, and (Right) Top 5 Neighborhoods w/Lowest Premium

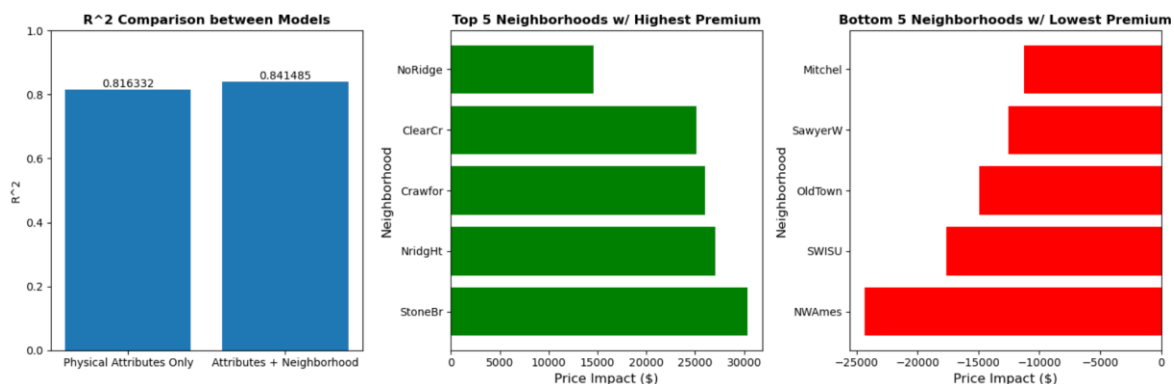
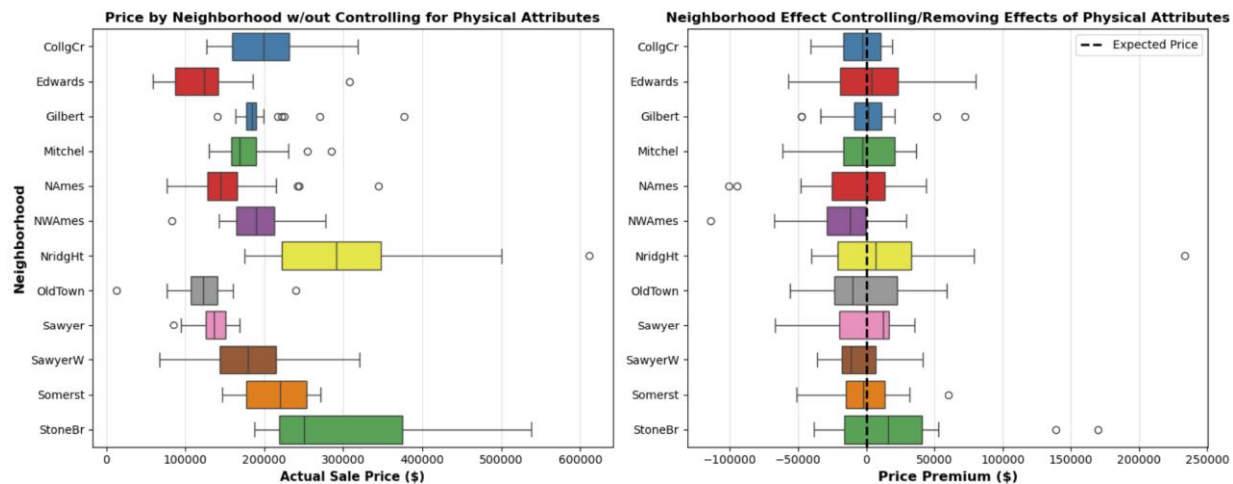


Figure 7 is a comparison of neighborhood price distributions before (Left) and after (Right) controlling for physical attributes. The left boxplot shows actual sale prices by neighborhood and has a large spread of prices. Neighborhoods like StoneBr and NridgeHt cluster at higher prices

exceeding \$300k while OldTown and Edwards are much lower. This spread is primarily due to the differences in home size and quality and not the neighborhood location itself.

The right boxplot in **Figure 7** plots price residuals after controlling for physical features: overall quality, living area, basement size, garage capacity, year built, and year remodeled. Positive residuals indicate homes sold above what their physical features predict, suggesting a neighborhood/location premium. The dashed vertical line at zero represents the expected price given physical features alone. General visual interpretation of this boxplot shows neighborhoods have a tight cluster around zero with most neighborhoods having $\pm \$25k$ location effect after removing physical feature contributions.

Figure 7: Neighborhood Prices Before (Left) / After (Right) Controlling for Physical Features



Assumptions Met and Verified

The assumptions for Model 2 were verified. The ANOVA F-test depends on the same regression assumptions outlined in Question 1 in terms of linearity and normality of residuals.

- Linearity: Verified in Question 1.
- Normality (**Figure 8**): Met and verified using Q-Q plot of residuals. Residuals were normally distributed.
- Constant Variance (**Figure 9**): Plotted residuals vs. y to show minimal fanning or funneling.
- Mean of 0 (**Figure 9**): Confirmed residuals were not uniformly above 0 or uniformly below 0 for any subsection.
- Independence (**Figure 9**): Plot of residuals vs. y did not indicate any clear patterns. Each home sale represents a unique home sale with no repeated or duplicate rows.
- Outliers: Similar to Model 1, the relative percentage of outliers did not significantly impact the results.
- Confirmed neighborhoods were mapped to the dummy variables. 19 variables were mapped correctly.
- Validated the sample size of all neighborhoods as set in the code to ensure all neighborhoods had at least 5 observations (min was 7 houses, max was 69 houses).

Figure 8: Q-Q Plot of Residuals – Model 2 w/Neighborhood

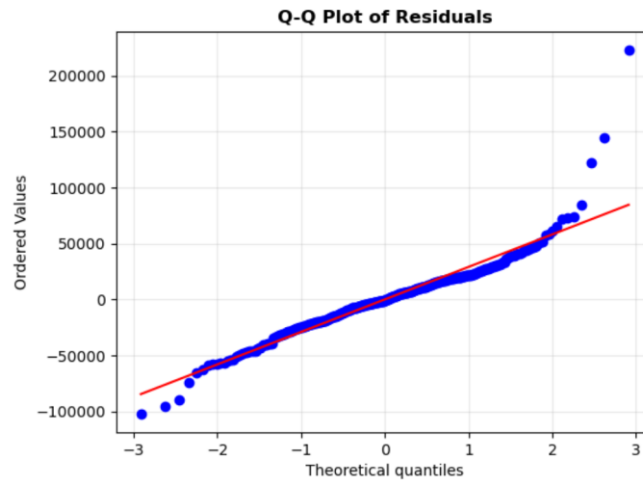
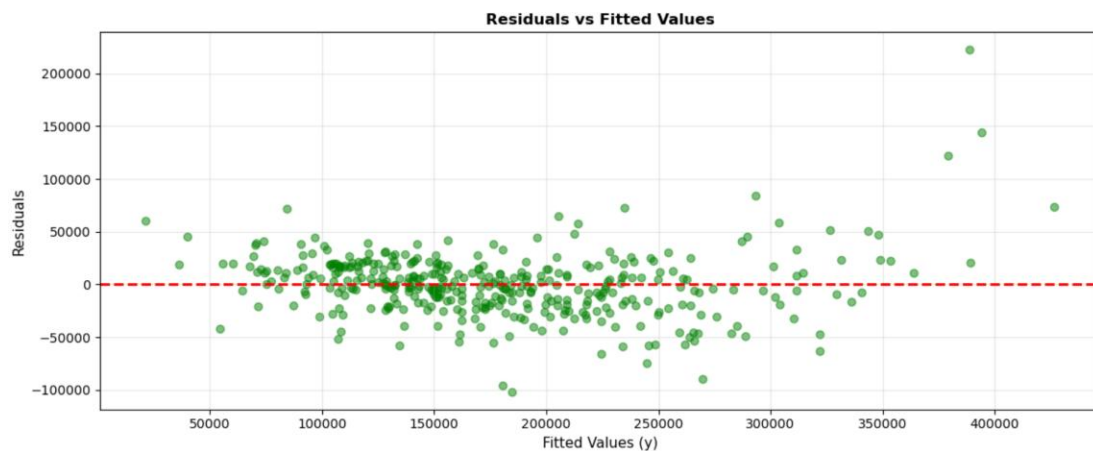


Figure 9: Constant Variance and Mean of 0: Residuals vs. Fitted



The ANOVA F-test confirms that neighborhood significantly affects price after controlling for physical features with $F = 3.015$ and $p = 0.000029 < 0.05$, so we reject H_0 . Although statistically significant when using the F-test, the overall neighborhood effect relative to the top correlated physical features is small. Physical attributes remain the primary driver of price, explaining 82% of the variance.

Research Question 3

Can it be predicted whether a house will sell above the median price based on its features?

A logistic regression model was used to classify homes to evaluate this question. The classification analysis shows that physical features can be used to accurately predict whether a home sells above or below median price in Ames, IA. The model achieves a 90.25% accuracy classifying homes as above or below the median of \$160k. It correctly predicts 361 of 400 homes.

Table 7: Classification Metrics

Metric	Value
Median Home Price	\$160,125
Classification Accuracy	90.25%
Correctly Classified Homes	361 of 400
Misclassified Homes	39
True Negative (correctly predicted below median)	178
True Positive (correctly predicted above median)	183
False Positive	22
False Negative	17

A normal approximation to a binomial test for classification accuracy using a Z-test evaluated whether the model performs better than random chance. The Z-statistic of 16.1 and p-value < 0.0001 indicate that the model performs significantly better than random chance in predicting whether a home will sell above or below the median price. A 95% confidence interval calculated for model accuracy shows a lower bound of 87.34% and an upper bound of 93.16%. This shows that even at the lower interval, the model is greater than random chance.

Table 8: Binomial Z-Test Summary

Statistic	Value
H ₀	0.50 (random chance)
Z-Statistic	16.1
p-value	< 0.0001
95% Confidence Interval	87.34% - 93.16%

The confusion matrix displayed in **Figure 10 (Left)** indicates that out of the 400 homes, 178 below median homes were correctly classified and 183 above median homes were correctly classified. Only 39 homes were misclassified as either false positives or false negatives.

Figure 10 (Right) shows the normalized feature importance which addresses the scale differences by standardizing features before fitting the model. The original regression coefficients initially suggested that garage capacity was more impactful than living area, i.e. one car space addition had a relatively larger impact than one square foot change. This did not account for the fact that area varies across hundreds of square feet while Garage.Cars only range from 0 to 4. Once normalized, living area (area) was the most important predictor with a standardized coefficient of 1.7, followed by Year.Built, and Garage.Cars. This means that living area has the largest impact on the probability of selling above or below the median. Year.Remod.Add had the smallest effect.

This data can inform homeowners to prioritize improvements to higher importance features (quality upgrades and additional living area) over lower impact features. The analysis can also be used by appraisers as quality checks where they could flag cases where physical features suggest different prices compared to appraised prices.

Figure 10: Logistic Classification: (Left) Confusion Matrix and (Right) Feature Importance

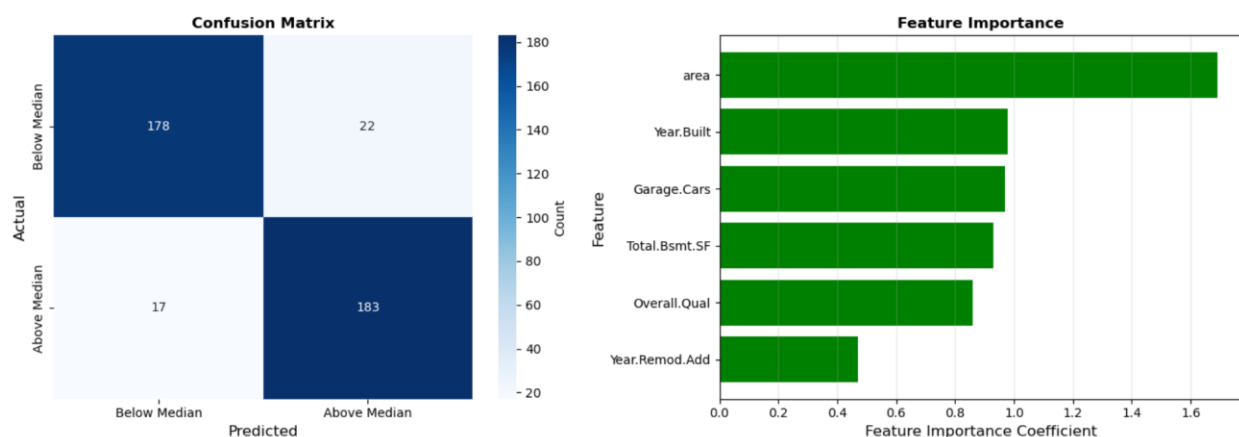
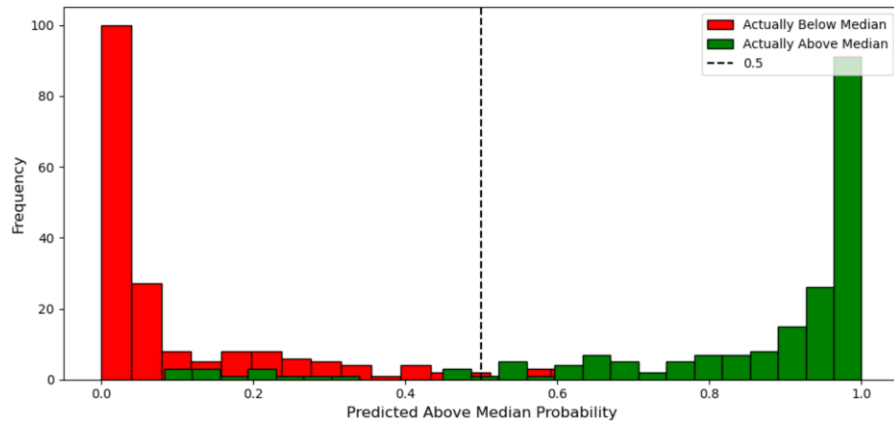


Figure 11 shows the distribution of predicted probabilities. Homes that sold below median price are shown in red and have a higher frequency near 0. Homes that sold above median are shown in green and have a higher frequency at 1. The clear separation between the high

frequency counts at 0 and 1 indicates the model can accurately predict homes selling above or below median price based on physical features.

Figure 11: Distribution of Predicted Probabilities



Assumptions Met and Verified

Additional assumptions were verified in Question 3 for the logistic regression model and the binomial test.

Logistic Regression Model:

- Confirmed the dependent variable (above median) contains only 0 and 1.
- Independence: Verified that each of the 400 predictions corresponds to a unique home sale with no duplicates.
- Sample Size: Confirmed an adequate sample size by looking at the total ratio of observation vs. predictors. There are 400 observations and 6 predictors which is a relatively high ratio of 66.7.
- Outliers: Verified minimal impact from outliers. Only a small percentage (3%) exceeded three standard deviations.

Binomial Test:

- Confirmed binary prediction is classified as correct (1) or incorrect (0).
- Verified each of the 400 predictions corresponds to a unique home sale.
- Sample Size: Validated normal approximation for Z-test by checking the sample size.
 - $n * p_0 = 200$ and $n * (1 - p_0) = 200$, both ≥ 10
- Checked assumption that $p_0 = 0.50$ for all predictions.

Based on the binomial test results above, the model accuracy of 90.25% predicts better than random chance with a p-value < 0.0001 . This indicates that the logistic regression model can accurately predict whether a home will sell above or below the median price based on its physical features at better than random chance, so we reject H_0 .

4. Conclusions

This analysis used several statistical methods to evaluate three practical research questions around property prices in the Ames, IA Housing Dataset. The statistical methods and tests are outlined in **Table 9**, and the key findings are summarized below.

Table 9: Statistical Methods and Tests Summary

Method	Research Question	How the Method was Applied	Results
Multiple Linear Regression	Q1	Predicting sale price using six physical features as the predictors	$R^2 = 0.82$ $F = 290$ $p < 0.0001$
ANOVA F-Test	Q2	Testing whether adding neighborhood to the physical features only model improves sale price prediction	$F = 3.01$ $p = 0.000029$
Logistic Regression	Q3	Binary (above/below median price) classification	90.25% Accuracy 361 of 400 Correctly Classified
Binomial Z-Test	Q3	Model accuracy evaluation	$Z = 16.1$ $p < 0.0001$
95% Confidence Interval	Q3	Model accuracy confidence interval	Lower CI: 87.34% Upper CI: 93.16%

Research Question 1: This question explores which measurable features have the strongest relationship with sale prices. Physical features including living area, overall quality, basement size, garage capacity, and year built/remodeled, are all strong predictors of sale price. A multiple linear regression model using these physical features explains 82% of the variance in sale price. The F-test and individual physical feature t-tests all showed statistical significance with $F = 290$ and $p < 0.0001$. Overall quality and living area showed the strongest impact on price.

Research Question 2: Neighborhood was shown to have a statistically significant impact on sale price when physical features are controlled. An ANOVA F-test confirmed this with $F = 3.01$ and p-value = 0.000029. However, neighborhood adds relatively small explanatory value of 2.5% beyond physical features. Neighborhood effects range from roughly -\$25k to +\$30k after controlling for physical features, but the primary variables that predict sale price are the home's physical features.

Research Question 3: A logistic regression model accurately classifies homes as above or below the median sale price of \$160k with 90.25% accuracy and a 95% confidence interval between 87.34% and 93.16%. A binomial Z-test confirmed this is more than random chance with a Z-statistic of 16.1 and p-value < 0.0001 . Living area was the most important predictor. This model could be used as a practical screening tool on other datasets for buyers or appraisers.

Overall, this analysis shows that a home's physical features are the strongest predictors of sale price in Ames. While neighborhood has a statistically significant effect, the contribution is relatively small at only 2.5% beyond what physical features explain. The logistic regression model validates the correlation of a home's physical features to price by classifying homes above or below median price with 90% accuracy. This approach could be applied to other housing markets to help buyers and sellers make more informed pricing decisions based on measurable home features.

5. Appendix

Python Script

The full Python script used to perform data cleaning, statistical analysis, hypothesis testing, and visuals generation is included as a separate file (Final_Project_Python.py).

Supporting Files

1. Ames_Dataset.csv – The original Ames Housing Dataset.
<https://www.openintro.org/data/index.php?data=ames>
2. Ames_Clean_Data.csv – Cleaned dataset exported by the Python script.

References

1. De Cock, Dean. "Ames, Iowa: Alternative to the Boston housing data as an end of semester regression project." *Journal of Statistics Education* 19.3 (2011).
2. [Scikit-learn: Machine Learning in Python](#), Pedregosa *et al.*, JMLR 12, pp. 2825-2830, 2011.
3. Harris, C.R., Millman, K.J., van der Walt, S.J. et al. *Array programming with NumPy*. *Nature* 585, 357–362 (2020). DOI: [10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2).
4. [J. D. Hunter, "Matplotlib: A 2D Graphics Environment", *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90-95, 2007.](#)
5. Waskom, M. L., (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021, <https://doi.org/10.21105/joss.03021>.